“Exploratory Data Analysis of Superstore Dataset Using Correlation and Pairplots”

A PROJECT SUBMITTED TO

SYNTECXHUB Technologies

Kanpur Nagar, Uttar Pradesh

FOR THE INTERNSHIP

IN

DATA SCIENCE

By

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Introduction

In today's highly competitive and data-driven business environment, organizations must rely on data analysis to make informed decisions and stay ahead in the market. Retail businesses, in particular, generate large amounts of transactional data and analyzing this data is essential to understand customer behaviour, product performance and overall profitability.

This project focuses on analyzing the Superstore dataset to study the relationships between key business variables such as Sales, Profit, Quantity and Discount. Understanding these relationships is important because factors like excessive discounts or low sales efficiency can directly impact a company's profit.

The need for this analysis arises from the current business situation where companies are facing intense competition, changing customer demands and pressure to maximize profits while maintaining sales. By using techniques such as correlation analysis, heatmaps and pairplots, this project helps identify important patterns and trends in the data.

The insights gained from this study can support better decision-making, such as improving pricing strategies, controlling discounts and focusing on factors that contribute most to profit. Overall, this project highlights the importance of data-driven analysis in improving business performance and achieving sustainable growth.

Objectives:

- To load the dataset and select important numerical variables using Pandas for analysis.
- To find the correlation between numerical variables and visualize it using a heatmap.
- To analyze pairwise relationships using pairplots (scatter matrix).
- To identify and summarize the strongest positive and negative relationships.

Methodology:

Data Source:
The dataset used for this project is the Superstore Dataset, downloaded from Kaggle. It contains 9,994 records (rows) and 21 columns, including information about sales transactions, customer details, product categories and key variables such as Sales, Quantity, Discount and Profit.

Variables Description:

- Sales: Total revenue generated from products
- Profit: Profit or loss earned from sales
- Quantity: Number of items sold
- Discount: Discount given on products

Process:
First, the dataset was loaded using the Pandas library and basic checks were performed to understand the data. Then, important numerical variables (Sales, Profit, Quantity, Discount) were selected for analysis.

Next, the correlation between these variables was calculated and visualized using a heatmap. A pairplot was also created to study the relationships between variables. Finally, the strongest positive and negative correlations were identified and interpreted to draw meaningful conclusions.

Data Analysis

- **To load the dataset and select important numerical variables using Pandas for analysis.**

Code:

```
import pandas as pd

# Load dataset

df = pd.read_csv("C:\\Users\\ACER\\Downloads\\Superstore Dataset.csv", encoding='latin1')

# Check dataset info

print(df.info())

# Select important numerical columns

numeric_df = df[['Sales', 'Profit', 'Quantity', 'Discount']]

print(numeric_df.head())
```

Output:

Dataset Info:

[5 rows x 21 columns]

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 9994 entries, 0 to 9993

Data columns (total 21 columns):

	Column	Non-Null Count	Dtype
0	Row ID	9994 non-null	int64
1	Order ID	9994 non-null	object
2	Order Date	9994 non-null	object
3	Ship Date	9994 non-null	object
4	Ship Mode	9994 non-null	object
5	Customer ID	9994 non-null	object
6	Customer Name	9994 non-null	object
7	Segment	9994 non-null	object
8	Country	9994 non-null	object
9	City	9994 non-null	object
10	State	9994 non-null	object
11	Postal Code	9994 non-null	int64
12	Region	9994 non-null	object
13	Product ID	9994 non-null	object
14	Category	9994 non-null	object
15	Sub-Category	9994 non-null	object
16	Product Name	9994 non-null	object
17	Sales	9994 non-null	float64
18	Quantity	9994 non-null	int64

19	Discount	9994 non-null	float64
20	Profit	9994 non-null	float64

dtypes: float64(3), int64(3), object(15)

memory usage: 1.6+ MB

None

Important Numerical Columns:

	Sales	Profit	Quantity	Discount
0	261.9600	41.9136	2	0.00
1	731.9400	219.5820	3	0.00
2	14.6200	6.8714	2	0.00
3	957.5775	-383.0310	5	0.45
4	22.3680	2.5164	2	0.20

- **To find the correlation between numerical variables and visualize it using a heatmap**

Code:

```
import seaborn as sns

import matplotlib.pyplot as plt

import numpy as np


# Compute correlation matrix

corr_matrix = numeric_df.corr()


print("Correlation Matrix:")

print(corr_matrix)


# Create mask for upper triangle

mask = np.triu(np.ones_like(corr_matrix, dtype=bool))


# Plot heatmap with mask

plt.figure(figsize=(10, 8))

sns.heatmap(corr_matrix,

        mask=mask,

        annot=True,
```

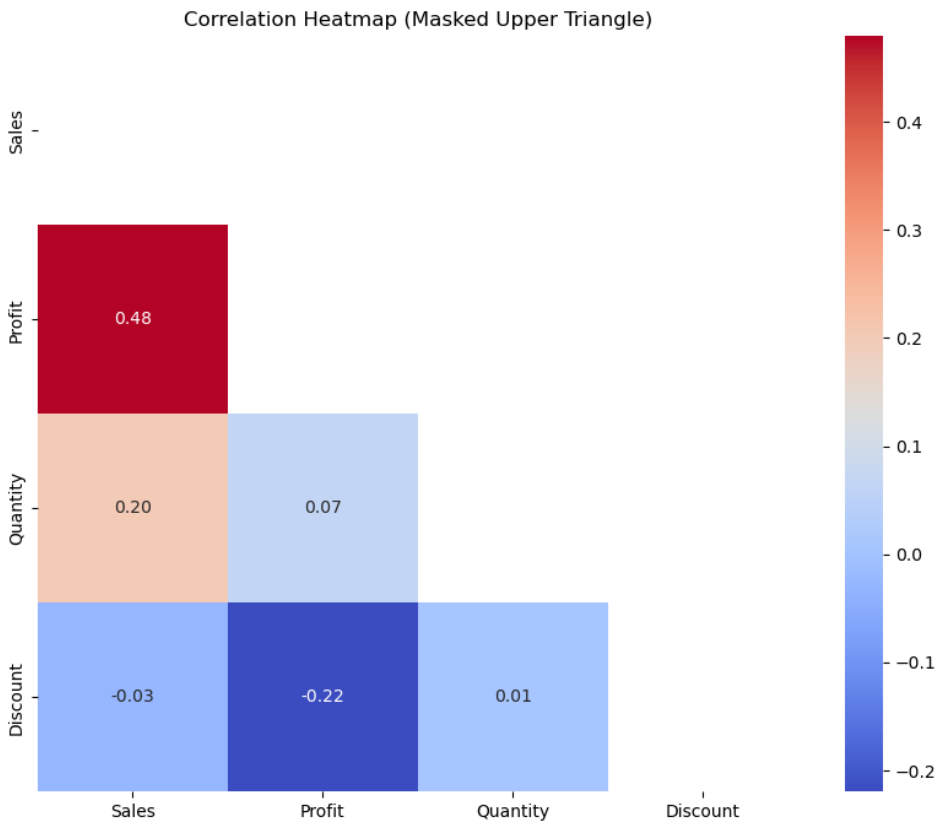
```
        cmap='coolwarm',
        fmt=".2f")

plt.title("Correlation Heatmap (Masked Upper Triangle)")
plt.show()
```

Output:

Correlation Matrix:

	Sales	Profit	Quantity	Discount
Sales	1.000000	0.479064	0.200795	-0.028190
Profit	0.479064	1.000000	0.066253	-0.219487
Quantity	0.200795	0.066253	1.000000	0.008623
Discount	-0.028190	-0.219487	0.008623	1.000000



Interpretation:
The heatmap shows that Sales and Profit have a moderate positive correlation (0.48), meaning higher sales lead to higher profit. Quantity has a weak positive relationship with Sales (0.20) and Profit (0.07). Discount has a negative correlation with Profit (-0.22) and almost no relationship with Sales (-0.03).

Conclusion:

From this analysis, we can say that earning more profit mainly depends on increasing sales, but giving too many discounts can reduce profit. Selling more items does not always guarantee higher profit. So, the business should focus on improving sales while keeping discounts under control to maintain good profit.

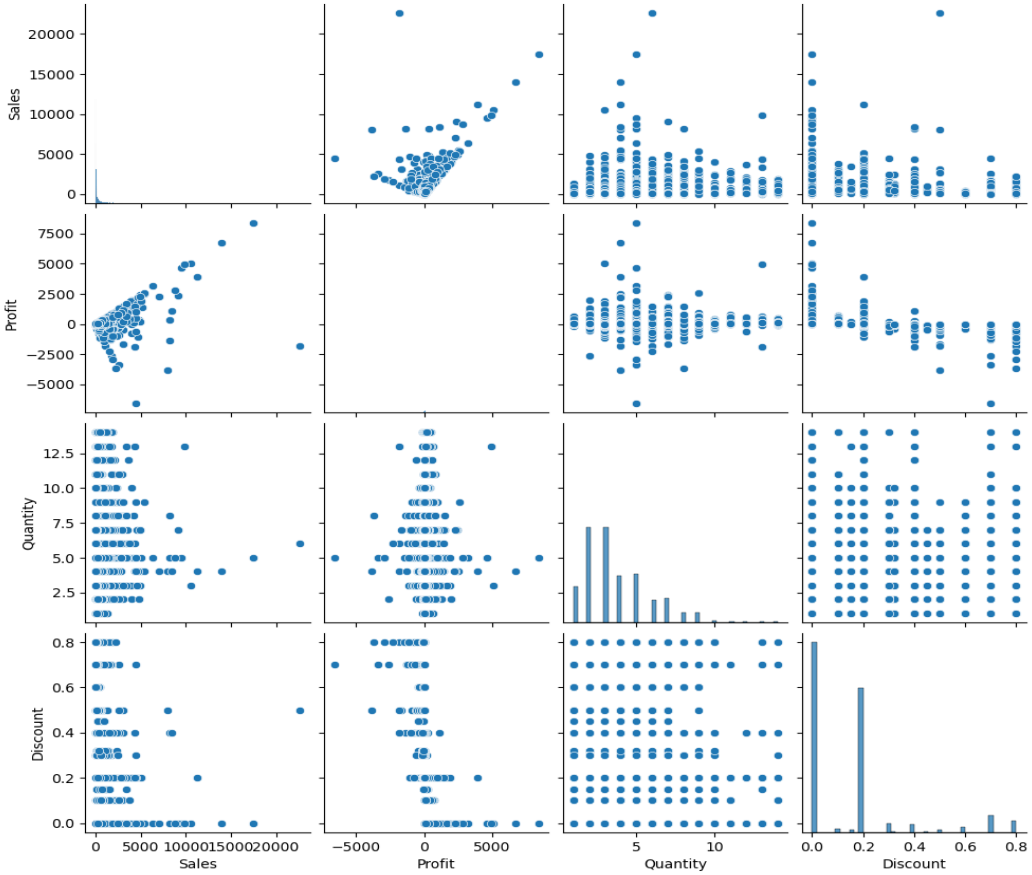
- **To analyze pairwise relationships using pairplots (scatter matrix).**

Code:

```
sns.pairplot(numeric_df)

plt.show()
```

Output:



Interpretation:

The pairplot shows that Sales and Profit follow an upward trend, indicating a positive relationship. Profit decreases as Discount increases, showing a negative trend. Quantity does not show a strong pattern with Sales or Profit, as the points are scattered.

Conclusion:

Higher sales usually bring higher profit, but giving more discounts can reduce profit. Selling more items does not always increase profit, so the focus should be on smart pricing and controlled discounts.

- **To identify and summarize the strongest positive and negative relationships.**

Code:

```
corr_unstacked = corr_matrix.unstack()

# Remove self-correlation
corr_unstacked = corr_unstacked[corr_unstacked != 1]

# Sort values
sorted_corr = corr_unstacked.sort_values()

print("Strongest Negative Correlations:")
print(sorted_corr.head(5))

print("\nStrongest Positive Correlations:")
print(sorted_corr.tail(5))
```

Output:

Strongest Negative Correlations:

Profit	Discount	-0.219487
Sales	Discount	-0.028190
Quantity	Discount	0.008623

Strongest Positive Correlations:

Quantity	Profit	0.066253
Sales	Quantity	0.200795
Sales	Profit	0.479064

Conclusion:

The strongest positive relationship is between Sales and Profit (0.48), meaning higher sales lead to higher profit. The strongest negative relationship is between Discount and Profit (-0.22), showing that higher discounts reduce profit. Other relationships are weak, so the main focus should be on increasing sales and controlling discounts to maintain profit.

Overall Conclusion:

This project analyzed the Superstore dataset to understand how key variables like Sales, Profit, Quantity and Discount are related. The analysis shows that Sales has the strongest positive impact on Profit, meaning increasing sales generally improves profitability. However, Discount has a negative effect on Profit, indicating that offering high discounts can reduce overall earnings. Quantity was found to have only a weak influence on both Sales and Profit.

From a business perspective, the results suggest that companies should focus on increasing sales while carefully managing discount strategies. Simply selling more items or giving heavy discounts does not guarantee higher profit. Therefore, adopting a balanced approach in pricing and sales strategies can help improve overall business performance and profitability.